Next-Day Air Quality (AQI) Forecasting for Indian Cities

Problem Statement · DataTalks.Club ML Zoomcamp Midterm Project

1. Background: What is AQI?

The Air Quality Index (AQI) is a single number that tells the public how polluted the air is and what health risks it carries. India's National AQI, launched by the Central Pollution Control Board (CPCB) in 2014, combines up to eight pollutants: $PM_{2.5}$, PM_{10}, NO_2, SO_2, CO, O_3, NH_3, and lead. It reports the worst of their sub-indices as the overall AQI.

The index is grouped into six categories:

AQI Range	Category	Health Impact
0–50	**Good**	Minimal impact
51–100	**Satisfactory**	Minor breathing discomfort for sensitive people
101–200	**Moderate**	Discomfort for people with lung or heart disease, children, and older adults
201–300	**Poor**	Breathing discomfort for most people on prolonged exposure
301–400	**Very Poor**	Respiratory illness on prolonged exposure
401–500	**Severe**	Affects healthy people and seriously harms those with existing diseases

2. The Problem: Air Pollution in India

India has some of the most polluted cities in the world. Many cities in the Indo-Gangetic Plain, including Delhi, Ghaziabad, Noida, Lucknow, Patna, and Kanpur, regularly cross into the “Poor,” “Very Poor,” and “Severe” categories, especially during winter.

Main causes of pollution

- Vehicle emissions in dense, fast-growing cities
- Industrial activity and coal-based power plants
- Construction and road dust, a major source of PM_{10}
- Crop residue (stubble) burning in Punjab and Haryana during October–November
- Household burning of biomass, wood, and waste
- Firecrackers during festivals
- Winter weather, when low wind speeds, temperature inversion, and low mixing height trap pollutants close to the ground

3. Impact

Health

- Fine particulate matter ($PM_{2.5}$) is small enough to enter the lungs and bloodstream. It is linked to asthma, chronic obstructive pulmonary disease (COPD), lung cancer, heart attacks, strokes, and poor lung development in children.

- A 2020 study in *The Lancet Planetary Health* attributed about 1.67 million deaths in India in 2019 to air pollution, roughly 18% of all deaths that year.
- Children, older adults, pregnant women, outdoor workers, and people with existing heart or lung conditions are the most vulnerable.

Economic

- The same Lancet study estimated that lost output from pollution-related illness and early deaths cost India about 1.4% of its GDP in 2019.
- Pollution also increases healthcare spending, reduces worker productivity, and lowers tourism.

Social and daily life

- Schools in Delhi NCR have had to close or move online during severe pollution episodes.
- Construction bans and vehicle restrictions under Delhi's Graded Response Action Plan (GRAP) disrupt work and income, especially for daily-wage workers.
- Flights and road travel are affected by smog and low visibility.

4. The Gap: Why Forecasting Matters

Current AQI dashboards, such as CPCB's SAMEER app, mostly report what the air quality is right now. By the time AQI reaches “Severe,” people have already been exposed and authorities are reacting rather than preventing.

A reliable next-day forecast would allow:

- **Citizens** to plan outdoor activities, exercise, and commutes, and to take precautions such as masks and air purifiers
- **Schools** to decide on closures or indoor activities in advance
- **Hospitals** to prepare for more respiratory and cardiac cases
- **Local authorities** to trigger GRAP measures, such as construction bans, traffic restrictions, and water sprinkling, before pollution peaks rather than after

5. Project Objective

This project builds a machine learning model that predicts the **next day's AQI category** for an Indian city using:

- **Air quality history:** today's AQI, its change since yesterday, its 7-day mean and spread, and how far today is from the previous week's mean
- **Prominent pollutants:** flags for the pollutants driving the AQI each day ($PM_{2.5}$, PM_{10}, O_3, CO, NO_2, SO_2), and a flag for days whose AQI was filled from the day before (gaps of up to 3 days)
- **Weather:** daily temperature (mean, max, min, range), humidity, dew point, rainfall, surface pressure, cloud cover, wind speed, and wind direction as east-west and north-south parts
- **Calendar:** the month, which captures winter and the October–November stubble-burning period
- **The city**

The task is framed as a **multi-class classification** problem whose target is one of the six AQI categories on the following day.

Why weather matters: the same emissions can produce very different AQI depending on the weather. Low wind, high humidity, cold nights, and a small daily temperature range (signs of a temperature inversion) trap pollutants near the ground, while rain and strong wind clear them.

Weather features let the model anticipate these conditions.

> **Why not predict today's AQI?** AQI is computed directly from pollutant readings, so predicting the current AQI from current pollutants would only re-learn a known formula. Forecasting the next day's AQI is a genuine prediction problem with real practical value.

6. Dataset

The modelling dataset combines two free, public sources into one row per city per day:

- **CPCB daily AQI bulletins (2015–2025)**, compiled by Urban Emissions in the *AQI_bulletins* GitHub repository (GPL-3.0). Each bulletin gives a city's AQI value, AQI category, prominent pollutants, and number of monitoring stations.
- **Open-Meteo historical weather archive**, downloaded hourly for each city's coordinates and aggregated to daily values.

It covers **14 cities** from May 2015 to December 2025, about 51,000 city-days:

Region	Cities
Delhi NCR	Delhi, Noida, Greater Noida, Ghaziabad, Gurugram, Faridabad
Indo-Gangetic Plain	Lucknow, Kanpur, Patna
Other metros (cleaner air, for contrast)	Mumbai, Kolkata, Chennai, Bengaluru, Hyderabad

Each city has a complete daily calendar, so days without a bulletin (about 4%) appear as missing rows rather than disappearing. This keeps the trend features honest: the change “since yesterday” really is since yesterday.

Train, validation, and test split. The data is split by time rather than at random, so the model is always evaluated on days after the ones it learned from. After preparation there are 48,632 city-days:

- **Model development:** train on 2015–2022 (33,543 rows) and validate on 2023 (5,104 rows) to tune and select models, confirmed with **rolling validation over 2019–2023**: five years, each scored by a model trained only on the years before it
- **Final model:** retrained on 2015–2024 (43,755 rows) with the settings chosen above
- **Test:** 2025 (4,877 rows), used **once** for the final evaluation

Limitation: the bulletins report the overall AQI and the names of the prominent pollutants, but not individual pollutant concentrations. Recent AQI history stands in for pollutant levels, since AQI is the highest of the pollutant sub-indices.

7. Success Criteria

Baseline. The model should outperform a simple **persistence baseline**, which assumes tomorrow's AQI category equals today's. On the 2025 test year this baseline scores a **macro F1 of 0.659** and an accuracy of 0.733 (macro F1 0.634 over rolling validation 2019–2023). The model should beat it in every validation year, not only on average.

Evaluation metrics. The classes are imbalanced: Severe is 2.4% of days, and Very Poor or Severe ("Dangerous") 11.8%. A model that never forecast Severe would still score well on accuracy, so the metrics focus on precision and recall:

Metric	What it measures	Why
Macro F1 (main metric)	F1 (precision and recall combined) for each category, averaged so the six count equally	Rare categories weigh as much as common ones; used to compare and choose models
Dangerous recall	Share of Very Poor or Severe days that were forecast as such	A missed dangerous day is the costly error
Dangerous precision	Share of Very Poor or Severe forecasts that were right	Too many false alarms and people stop trusting the alert
Dangerous PR AUC	How well the predicted probability of a dangerous day ranks the risky days, at every threshold	Users can set their own alert threshold; guessing scores the share of dangerous days, not 0.5

Precision, recall, and F1 for each category and a confusion matrix are reported as well. Accuracy and ROC AUC are given for comparison only, since both are dominated by the many ordinary days.

Deployment. The final model is deployed as a **FastAPI web service in Docker**, hosted in the cloud. For a city it returns the predicted category, the probability of every category, the probability of a Severe and of a Dangerous day, and an alert whose threshold each user can set. It can also build its own input from the latest CPCB bulletin and weather.